Scalar subspaces and unbounded algebraic local rank

Human Compression project

Abstract

The algebraic residue convention in the fixed rank-three Suzuki article admits locally doubly transitive, residually weakly primitive pregeometries of arbitrarily large rank. A coordinate subspace supplies the soluble parabolic; deleting its coordinates matches intersections of independently chosen maximal subfields. This refutes the article's separate universal rank assertion. No claim about full incidence residues in higher rank is needed.

1 The assertion and acquired structure

The introduction of D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63, asserts on printed pp. 44–45 that no pregeometry of rank greater than three satisfies both residual weak primitivity and local double transitivity, and that rank three occurs only when $q-1$ is prime. These also follow from the complete list stated in Theorem 6.1. We use the definitions on pp. 45–46: the residue at a flag of the distinguished chamber is formed from the parabolic intersections. The local condition is imposed on these algebraic rank-one residues, not on all elements in an actual incidence panel. This definition applies to every finite type set; it is not restricted to rank three. Flag transitivity is not assumed.

We retain the classical faithful Suzuki coordinates and subgroup theorem. The standard group over a subfield consists of matrices with parameters in that subfield. Its point stabilizer is $B_t = U_t H_t$, of order $t^2(t-1)$. The standard involution centre is $Z_F = \{u(0,b) : b \in F\}$, with addition in its parameter. It centralizes U_t, and H_t acts by the full scalar group $\mathbb{F}_t^*$. Indeed the standard action is $b \mapsto k^{\theta+2}b$ and $\gcd(\theta+2, t-1) = 1$. These identities and the overgroup correspondence below are acquired from the accepted scalar-subspace and subfield-family notes. A simple subfield group is maximal at prime relative field degree. The action of $\mathrm{Sz}(t)$ on $\mathrm{Sz}(t)/B_t$ is its classical doubly transitive action on t^2+1 points.

2 Independent field coordinates

Let $t = 2^e \geq 8$, with e odd. Choose distinct odd primes $a_1, \ldots, a_k$, with $k \geq 2$, and put

$$A = \prod_{i=1}^{k} a_i, \quad F = \mathbb{F}_{t^A}, \quad K = \mathbb{F}_t, \quad F_i = \mathbb{F}_{t^{A/a_i}}.$$

Choose $x_i \in \mathbb{F}_{t^{a_i}} \setminus K$, and set

$$T = K + Kx_1 + \cdots + Kx_k, \qquad T_J = K + \sum_{i \notin J} Kx_i \quad (J \subseteq \{1, \ldots, k\}).$$

Lemma 1. *The vectors $1, x_1, \ldots, x_k$ are linearly independent over K. Moreover*

$$F_J := \bigcap_{j \in J} F_j = \mathbb{F}_{t^{A/\prod_{j\in J} a_j}}, \qquad T \cap F_J = T_J,$$

where the empty intersection means F.

Proof. The field intersection formula follows from the gcd of the degrees. The automorphism $\sigma_i : y \mapsto y^{t^{A/a_i}}$ fixes F_i and all x_j for $j \neq i$. Its restriction to $\mathbb{F}_{t^{a_i}}$ has fixed field K, since a_i is coprime to A/a_i. Hence $\sigma_i(x_i) \neq x_i$. Applying $\sigma_i - 1$ to a relation $c_0 + \sum c_j x_j = 0$, with $c_j \in K$, gives $c_i(\sigma_i(x_i) - x_i) = 0$. Thus every $c_i = 0$, and then $c_0 = 0$. A vector in $T \cap F_i$ must likewise have zero x_i coefficient. Imposing this for all $i \in J$ proves the second equality. □

For a K-subspace $W \supseteq K$ of F, write

$$Q_W = Z_W B_t, \qquad Z_W = \{u(0, b) : b \in W\}.$$

This is a group: U_t centralizes Z_W, and H_t preserves W. Its order is $t|W|(t-1)$, since $Z_W \cap B_t = Z_K$ has order t. Every overgroup of Q_V inside Q_W is Q_D for a K-subspace $V \leq D \leq W$. Indeed every element has form zb, with $z \in Z_W$, $b \in B_t$. A subgroup containing B_t is therefore of this form with an additive subgroup of W containing K. Normalization by H_t makes that subgroup a K-subspace. Consequently

$$Q_V \text{ is maximal in } Q_W \iff \dim_K(W/V) = 1. \tag{1}$$

For any intermediate field $E \supseteq K$, unique coordinates give

$$\mathrm{Sz}(E) \cap Q_W = Q_{W \cap E}. \tag{2}$$

In fact multiplying by an element of $B_t \leq \mathrm{Sz}(E)$ reduces membership to the central parameter in $W \cap E$.

3 The construction and every residue check

Theorem 2. *In $G = \mathrm{Sz}(t^A)$ take the $k+1$ parabolics*

$$G_i = \mathrm{Sz}(F_i) \quad (1 \leq i \leq k), \qquad G_* = Q_T.$$

Their coset pregeometry satisfies the source's residual weak primitivity and algebraic local double transitivity conditions. Its rank is $k+1$. For $k = 2$ it is an incidence geometry of rank three, and $q-1$ is composite. Thus the source's general rank bound and its restriction of rank three to prime $q-1$ are false.

Proof. For $J \subseteq \{1, \ldots, k\}$, unique field coordinates, Lemma 1, and (2) give all intersections:

$$G_J = \mathrm{Sz}(F_J), \qquad G_{J \cup \{*\}} = Q_{T_J}. \tag{3}$$

The intersection of all parabolics is $B_t = Q_K$. Every parabolic is proper: each G_i is a prime-degree subfield, and Q_T is contained in the ambient point stabilizer. The intersections in (3) decrease strictly when any type is added.

We verify residual weak primitivity for every proper type subset. If the subset omits $*$ and some i, adjoining i is a prime-degree simple subfield inclusion, hence maximal. If it contains all $1, \ldots, k$ and omits only $*$, its group is $\mathrm{Sz}(t)$ and adjoining $*$ gives its maximal point stabilizer B_t. If the subset contains $*$ and omits an i, adjoining i removes exactly the independent coordinate Kx_i from T_J. Equation (1) gives maximality. This exhausts every proper subset. The last index is t, not a prime; maximality follows from scalar irreducibility as in (1).

The rank-one algebraic residue obtained by omitting $*$ is the coset action of $\mathrm{Sz}(t)$ on $\mathrm{Sz}(t)/B_t$, doubly transitive by the classical action. Omitting i gives Q_{K+Kx_i} on its cosets of B_t. These cosets identify with $(K + Kx_i)/K \cong K$. The centre provides all translations, while

H_t provides every nonzero scalar multiplication. Its induced action is therefore the affine group $K \rtimes K^*$, doubly transitive on t points. These are all the source's required local actions.

For $k = 2$, every rank-three coset pregeometry is a geometry: an incident pair has a common representative, which supplies the third coset and extends it to a chamber. Singletons extend in the same way. No flag transitivity follows from this observation. Finally $q - 1 = (t-1)(1+t+\cdots+t^{A-1})$ has both factors greater than one. The rank-three example thus has composite $q - 1$. Since k is arbitrary, (3) gives pregeometries of arbitrarily large rank. □

Concrete parameters are $t = 8$, $(a_1, a_2) = (3, 5)$, giving rank three at $q = 2^{45}$; and $t = 8$, $(a_1, a_2, a_3) = (3, 5, 7)$, giving rank four at $q = 2^{315}$. These are symbolic examples: any x_i in its displayed subfield outside K suffices. No ambient Suzuki group is enumerated.

Attribution and scope. Suzuki structure and its natural doubly transitive action are classical premises. The correspondence $W \mapsto Q_W$ and its maximality criterion reuse the accepted scalar-subspace theory. The contribution here is the independent field-coordinate construction and its application to the source's separate universal rank claim. No full incidence thinness, higher-rank geometry property, or flag transitivity is asserted. This result does not dispose of a different statement using full actual panels or additional geometric hypotheses. It is an additional correction within the already refuted original, not new manuscript or principal refutation credit. No computation, Lean declaration or compression factor is introduced.